

Data Privacy: Homeworks #3

Due on Wednesday 11PM, May 27, 2025

Note: Submit homework via LearnUS.

1. Show that the minimum mean square estimate is $\hat{X}(Y) = \mathbb{E}[X|Y]$. You can start from

$$\mathbb{E}[(\hat{X}(Y) - X)^2] = \mathbb{E}[(\hat{X}(Y) - \mathbb{E}[X|Y]) + (\mathbb{E}[X|Y] - X))^2]$$

2. Let X be a random variable and $Z \sim \mathcal{N}(0, \sigma^2)$ be an independent Gaussian noise. Let $Y = X + Z$ be noise corrupted version of X .

- (a) Show that

$$\mathbb{E}[X|Y = y] = y + \sigma^2 \frac{\partial}{\partial y} \log P_Y(y).$$

Hint: You can start from

$$P_Y(y) = \int P_X(x) P_Z(y - x) dx$$

and compare it with $\frac{\partial}{\partial y} P_Y(y)$.

- (b) (**Bonus**) What about $\text{Var}[X|Y = y]$?

3. Let q be the forwarding distribution of DDPM. Show that

$$q(X^{(n-1)}|X^{(n)}, X^{(0)}) = \mathcal{N}\left(\mu_n(X^{(n)}|X^{(0)}), \frac{1 - \bar{\alpha}_{n-1}}{1 - \bar{\alpha}_n} \beta_n\right)$$

where

$$\mu_n(X^{(n)}|X^{(0)}) = \frac{1}{\sqrt{1 - \beta_n}} \left(X^{(n)} + \beta_n \frac{\partial}{\partial X^{(n)}} \log q(X^{(n)}|X^{(0)}) \right)$$

Hint: Given $X^{(0)}$, all distributions are Gaussian. More precisely,

$$\begin{bmatrix} X^{(n-1)} \\ X^{(n)} \end{bmatrix} \Big| X^{(0)}$$

is multivariate Gaussian. You can use the fact that if

$$\begin{bmatrix} U \\ V \end{bmatrix} \sim \mathcal{N}\left(\begin{bmatrix} \mu_U \\ \mu_V \end{bmatrix}, \begin{bmatrix} \sigma_{UU} & \sigma_{UV} \\ \sigma_{VU} & \sigma_{VV} \end{bmatrix}\right)$$

then

$$U|V \sim \mathcal{N}(\mu_U + (\sigma_{UV}/\sigma_{VV})(V - \mu_V), \sigma_{UU} - \sigma_{UV}^2/\sigma_{VV}).$$

Also note that we have an analytic formula for

$$\frac{\partial}{\partial X^{(n)}} \log q(X^{(n)}|X^{(0)})$$

4. Show that

$$D_{\text{KL}}(\mathcal{N}(\mu_0, \sigma_0^2 I) \parallel \mathcal{N}(\mu_1, \sigma_1^2 I)) = \frac{1}{2\sigma_1^2} \|\mu_1 - \mu_0\|^2 + \frac{d}{2} \left(\frac{\sigma_0^2}{\sigma_1^2} - 1 \right) + d \log \frac{\sigma_1}{\sigma_0}.$$

where d is dimension of Gaussian distributions and $\mu_0, \mu_1 \in \mathbb{R}^d, \sigma_1, \sigma_2 > 0$.

5. Consider the DDIM forward process.

$$\begin{aligned} q(X^{(N)} | X^{(0)}) &= \mathcal{N}(\sqrt{\bar{\alpha}_N} X^{(0)}, (1 - \bar{\alpha}_N) I) \\ q(X^{(n)} | X^{(n+1)}, X^{(0)}) &= \mathcal{N}(\sqrt{\bar{\alpha}_n} X^{(0)} + \frac{\sqrt{1 - \bar{\alpha}_n - \sigma_{n+1}^2}}{\sqrt{1 - \bar{\alpha}_{n+1}}} (X^{(n+1)} - \sqrt{\bar{\alpha}_{n+1}} X^{(0)}), \sigma_{n+1}^2 I) \end{aligned} \quad (1)$$

for $\sigma_n \geq 0$. Show that equation 1 matches the marginal distribution of DDPM.

6. **Guidance.** In diffusion models, classifier-guided conditional generation uses

$$\nabla_x \log p_t(x) + \omega \nabla_x \log p_t(y | x)$$

with a classifier scale parameter $\omega \geq 1$. However, many papers in the literature add classifier guidance to an already conditional model via

$$\nabla_x \log p_t(x | y) + s \nabla_x \log p_t(y | x)$$

with $s \geq 0$. Show that the two are equivalent with $s = \omega - 1$.

7. **Langevin** Let $p(x)$ is a probability density function that is smooth and strictly positive for all $x \in \mathbb{R}^d$. Let $\{p_t\}_{t \in [0, T]}$ be the marginal density functions of the Langevin SDE

$$dX_t = \frac{1}{2} \nabla_{X_t} \log p(X_t) dt + dW_t.$$

Show that if $p_0 = p$, then $p_t = p$ for all $t > 0$.